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From Traffic-Flow Capacity to Recovery Capacity: An Episode-Based Framework for Multi-Sector Urban Resilience

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Abstract:	Urban systems experience disruptions that differ in magnitude, duration, and final recovery outcome. Existing resilience measures commonly emphasize performance loss, recovery duration, or trajectory area, but they do not directly represent the peak rate at which a disrupted system recovers. Inspired by the traffic-flow view of capacity as a dynamic, state-dependent peak rate, this paper develops an episode-based recovery-capacity framework. A Double Quadratic Queue (DQQ) procedure is used as an episode filter to identify critical transition points, and a Linear-Exponential Adaptation (LEA) model represents the recovery trajectory. Single-period recovery capacity is defined as the maximum healing rate, $R_{cap}=ab e^{-c}$, while multi-episode stress--strain relationships provide complementary descriptive measures of disruption sensitivity and transition loss or gain. Model comparison using Google COVID-19 Community Mobility Reports shows that LEA generally provides lower fitting and prediction errors than benchmark recovery models. Extreme-heat episodes in Phoenix and post-earthquake infrastructure restoration in Japan are included as transfer illustrations with case-specific variables and interpretations. The resulting framework is descriptive and diagnostic rather than causal, and it provides a focused connection between observed recovery episodes and dynamic capacity analysis.
Response to Reviewers:	

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Dear Editors of Transportation Research Part C,

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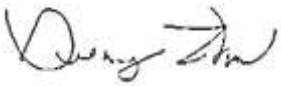
Please find enclosed our revised manuscript, “From Traffic-Flow Capacity to Recovery Capacity: An Episode-Based Framework for Multi-Sector Urban Resilience,” for consideration for publication in Transportation Research Part C.

We sincerely appreciate the opportunity to revise the manuscript and thank the editors and reviewers for their constructive comments. We have addressed all comments and provided a point-by-point response. The revisions have improved the focus, clarity, and presentation of the paper.

The manuscript is original, has not been published, and is not under consideration elsewhere. All authors have approved the submission, and there are no conflicts of interest to declare.

Thank you for your consideration.

Sincerely,



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Statement of Revision

Comments by Editor

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed.

We thank the editor and both reviewers. Read together, the two reviews identify one underlying problem: the previous manuscript combined several partially connected components—trajectory modeling, a thermal queue model, latent state-space dynamics, network reliability, and yield-point machinery—and the principal contribution was obscured. The revision therefore follows a *consolidation* strategy rather than a defense strategy. Components that could not be defended without a second paper's worth of methodological development have been removed, and the manuscript is now organized around a single contribution: recovery capacity defined as the peak healing rate, $R_{cap} = abe^{-c}$, of a bounded disruption–recovery episode, inspired by the traffic-flow view of capacity as a dynamic, state-dependent peak rate.

Concretely, the revision removes (i) the multi-sector state-space/SVD component and the Resilience Capacity Index; (ii) the series–parallel network-reliability calculation in the Japan case; (iii) the internal dissipation mechanism of the thermal queue model, retaining only the observed episode quantities D and P; and (iv) the offset-based unified yield-point definition, replaced by a descriptive regime-change diagnostic. The manuscript is reduced from 41 to 24 pages and from 41 to 14 displayed equation blocks, and the four capacity concepts of the previous version are reduced to two (R_{cap} and the descriptive stress–strain pair E, B). Section, figure, and table references below correspond to the revised manuscript. A colored change-tracked PDF comparing the previous submission with the revised manuscript accompanies this letter for version control.

Comments by Reviewer 1

Thank you for your submission. This paper innovatively extends traffic flow theory to the resilience analysis of urban multi-sectoral infrastructure ... However, to further enhance the theoretical rigor and logical coherence of the manuscript, I have the following queries and suggestions for your consideration during revision.

We thank Reviewer 1 for the positive evaluation and the constructive suggestions. Each point is addressed below.

[R1 - 1] In Section 2, the manuscript maps the Macroscopic Fundamental Diagram (MFD) and critical density from traffic flow theory onto the stress-strain relationship and yield point within the resilience model. While traffic systems are governed by explicit physical conservation laws, the recovery of socio-economic systems does not always adhere to strict conservation principles. It is recommended to explicitly discuss the boundary conditions of this theoretical mapping and the limitations of its physical interpretation in multi-sectoral scenarios.

We agree, and the revision addresses this comment in two ways. First, the mapping is now stated as phenomenological rather than as a transfer of physical laws: the Introduction characterizes the traffic-flow parallels as structural inspiration, and the subsection "Boundary Conditions and Scope of the Theoretical Mapping" (Section 2.4) delimits where the mapping holds—a meaningful baseline, a bounded disruption episode, and a measurable recovery trajectory—and restricts cross-sector comparison to normalized indicators or qualitative structural correspondence. Second, the scope of the mapping itself has been narrowed to what the paper actually uses. The subsection on network effects and queue-based integration has been deleted, together with the corresponding row of the mapping table, and the yield-point element of the analogy is now described only as a departure from an approximately linear fitted relationship, interpreted as a descriptive regime change rather than a material yield threshold. Table 1 retains the "Nature of Relationship" column (structural analogy, shared phenomenology, conceptual correspondence) and its caption states that no physical equivalence is asserted. See Sections 1, 2.3, and 2.4 and Table 1.

[R1 - 2] In the analogy involving Young's Modulus E, the model defines the "magnitude of disruption" as stress and "recovery duration" as strain. Given the disparate units of measurement across different sectors (e.g., traffic volume vs. thermal queuing), please clarify whether the derived E lacks a unified physical dimension and how this affects the model's comparability.

The reviewer is correct that E is case-specific in dimension and is not a universal scalar. Revised Section 3.1.3 defines E as the magnitude of the fitted stress–strain slope and states the dimensions of the fitted quantities explicitly. In the mobility analysis, stress and strain are relative deviations from baseline and the fitted E is dimensionless. In the thermal analysis, stress D has units of $\mathrm{W{\cdot}h/m^2}$ and strain $P - \bar{P}$ has units of hours, so E has units of $\mathrm{W/m^2}$, H has units of $\mathrm{W{\cdot}h/m^2}$, and $B = H/E$ has units of hours. The Japan application does not independently estimate E and is not used for numerical elasticity comparison. In addition, the comparability problem has been reduced at its source: the thermal application no longer defines a model-based capacity analogue (the former $R_{cap}^{(\mathrm{QTM})}$ and the associated modeled dissipation quantities have been removed), so no cross-dimensional capacity comparison is attempted anywhere in the paper. Raw parameter values are interpreted within case, and the manuscript states that E and B are case-specific comparative indicators rather than universal physical constants. See Sections 1 and 3.1.3.

[R1 - 3] Section 4.2 asserts that the LEA model outperforms traditional models (e.g., Weibull, Exponential) in fitting COVID-19 mobility data. Considering that the LEA model involves a higher number of parameters (a, b, c), its superior goodness-of-fit may stem from increased degrees of freedom. I suggest elaborating on the specific micro-mechanisms and physical meanings of these parameters in the

context of recovery dynamics, rather than relying solely on the reduction of Sum of Squares Error (SSE).

We agree that additional parameters remain a legitimate statistical concern even when they are interpretable, and the revision answers on both fronts. Interpretation: Section 3.1.1 gives one-sentence interpretations of each parameter—a sets the initial mobilization intensity ($h(t)/t \to a$ as $t \to 0^+$), b is the time of peak healing ($dh/dt = 0$ at $t^* = b$), and c is the dimensionless decay exponent controlling how sharply $h(t)$ falls from its peak—so the three parameters separate the intensity, timing, and shape of recovery, which two-parameter forms conflate. Evidence: model comparison is not based on in-sample SSE alone; the manuscript reports out-of-sample prediction error under a 70%/30% train–test split (Section 4.2, Figures 5h–i, with the prediction table as Table 4 in Section 5.1 and the state-level SSE summary as Table A2), aggregated across 50 states × 6 mobility types. The revised text also states explicitly that these are comparative predictive results, not significance tests. See Sections 3.1.1, 4.2, and 5.1.

[R1 - 4] Section 5 introduces Singular Value Decomposition (SVD) and state-space formulations to extract dimensionality-reduced recovery dynamics, proposing a Resilience Capacity Index (RCI). However, the analytical mapping between this macroscopic eigenvalue and the core microscopic parameters defined in Section 3 (e.g., initial recovery rate a and peak recovery time b) remains opaque. A direct mathematical or conceptual link between the two should be established.

We took this comment seriously, and it led to the largest structural change in the revision. The reviewer is correct that the mapping between the latent eigenvalue dynamics and the episode-level parameters (a, b, c) was opaque: the two constructs are different estimands—a finite-episode peak rate versus an asymptotic decay factor of a fitted linear system—and a rigorous mapping between them cannot be established without substantial additional assumptions and a separate identification and validation treatment. Rather than defend an under-developed link, we removed the entire state-space/SVD component, including the latent transition model, the eigenvalue and half-life interpretation, the nationwide spectral results, and the Resilience Capacity Index $RCI = 1 - \rho(A)$. The revised manuscript therefore contains a single estimand family—episode-level recovery capacity and cross-episode stress–strain response—and the mapping question no longer arises. The latent-dynamics direction is left as future work in the Conclusion, where it can receive the dedicated development it requires.

[R1 - 5] The study employs three highly divergent cases: COVID-19 mobility, extreme heat in Phoenix (via the QTM model), and post-earthquake recovery in Japan. Currently, these cases appear to independently validate distinct sub-models. I recommend adding a comprehensive discussion section to demonstrate how these heterogeneous perturbation types can be substantively compared and synthesized within the unified R_{cap} framework.

We agree with the premise of this comment, and in the revision we address it by consolidation rather than by an added synthesis section. The previous structure did invite the reading that three sub-models were validated independently, and the cross-case table introduced in our earlier revision compared quantities that were not commensurable. The revised manuscript instead applies one framework to all three cases: DQQ episode identification, the LEA trajectory with $R_{cap} = abe^{-c}$, and the descriptive stress–strain regression. The COVID-19 mobility case is the primary quantitative evaluation (calibration, model comparison, out-of-sample prediction); the Phoenix heat case and the Japan case are explicitly repositioned—in the Abstract, the Introduction, and Sections 4–5—as transfer illustrations that apply the same episode-based objects with case-specific variables and interpretations. Cross-case statements are limited to structural correspondence, dimensional quantities are interpreted within case, and no aggregate cross-case index is constructed. See the Abstract and Sections 1, 4, and 5.

[R1 - 6] The model relies extensively on Google COVID-19 Community Mobility Reports. It is crucial to note that these data represent percentage changes relative to a baseline period (Jan 3 – Feb 6, 2020), rather than absolute population or flow counts. Please supplement the discussion to address the potential impact of such relative data on the quantitative accuracy of absolute recovery capacity (R_{cap}), and

evaluate the model's robustness against non-stationary baselines.

We added a bounded discussion at the end of Section 4.1. Analytically, writing $Q(t) = N(t)/\bar{N} - 1$, the proportional healing rate satisfies $h_{\mathrm{rel}}(t) = \frac{1}{1+Q}\frac{dQ}{dt} = \frac{1}{N}\frac{dN}{dt} = h_{\mathrm{abs}}(t)$, so under a fixed multiplicative baseline, proportional rates—and hence R_{cap}—are invariant to rescaling. The COVID-19 application is accordingly interpreted as evidence on *relative* recovery structure rather than absolute demand recovery, and baseline non-stationarity is stated as an explicit limitation. Absolute quantities such as passengers or trips would require an external count-based reference, which the Japan service-availability case complements. See Section 4.1.

Comments by Reviewer 2

The paper proposes a recovery modelling framework for urban systems. While it addresses an interesting and relevant problem, the manuscript is difficult to follow. The narrative is fragmented ... A careful editorial restructuring of the paper is needed to improve coherence and readability.

We thank the reviewer for this candid assessment; it shaped the revision more than any other comment. The revision addresses the fragmentation at its source rather than through added transitions. Entire components have been removed: the multi-sector state-space/SVD section and the Resilience Capacity Index, the series–parallel network-reliability calculation, the internal mechanism of the thermal queue model, and the offset-based unified yield-point definition. The four research gaps of the previous version are consolidated to two, the four capacity concepts are reduced to two (R_{cap}; the descriptive pair E, B), and the abstract and title now state a single contribution. The manuscript falls from 41 to 24 pages and from 41 to 14 displayed equation blocks. The remaining structure follows one line—episode identification, recovery trajectory, peak capacity, cross-episode response, and three applications of the same objects—organized by the roadmap in Figure 1, to which the beginnings of the corresponding sections refer back. Each specific point is addressed below.

[R2 - 1] While the paper appropriately acknowledges Hani S. Mahmassani's contributions to traffic flow theory, the connection between his work and the proposed framework appears somewhat forced. Traffic flow theory is grounded in well-established physical conservation laws with clearly defined and measurable variables. In contrast, urban recovery processes involve heterogeneous agents, institutional constraints, governance structures, and path dependencies. The use of stress-strain analogies and the mapping from traffic capacity to "recovery capacity" is conceptually appealing, but it lacks a rigorous theoretical foundation. I suggest that the authors revise the introduction by toning down the link between the two directions.

This concern overlaps with Reviewer 1, Comment 1, and one coordinated revision addresses both. The connection has been both toned down and narrowed. The Introduction and Section 2 now characterize it as phenomenological—shared regularities of dynamic, state-dependent peak capacity and demand–supply adjustment—rather than as a transfer of conservation laws, and the framework is generalized through two core concepts instead of four: capacity as a state-dependent peak rate on a dynamic trajectory, and recovery as a demand–supply adjustment process shaped by behavioral learning. The subsection on network effects and queue-based integration has been deleted, together with the queue-dynamics row of the mapping table, so Section 2 claims only the parallels the paper actually uses. The boundary-conditions subsection (Section 2.4) specifies that recovery inflow–outflow relations are accounting relationships defined for a selected sector and observation period, and Table 1 retains the "Nature of Relationship" column with a caption stating that the mappings do not assert physical equivalence. The revised caption of Figure 2 likewise describes the arrows as conceptual parallels rather than universal equivalences. See Sections 1, 2.3, and 2.4, Table 1, and Figure 2.

[R2 - 2] The framework itself is largely descriptive and relies on curve fitting, yet the paper draws policy implications as if the model has causal explanatory power. For example, statements suggesting that regions with lower "healing capacity" should be prioritised for targeted interventions implicitly assume causal relationships that are not established by the observational data. This issue requires more careful discussion and clarification of the limits of the model's interpretability.

We agree, and the revision goes further than the earlier interpretive-scope paragraph. Section 6 opens by stating that the framework is descriptive and diagnostic rather than causal, that low estimated recovery capacity identifies trajectories warranting further investigation rather than interventions to be prescribed, and that the policy implications are questions for further evaluation via causal analysis, simulation, or quasi-experimental designs. In addition, the discussion passages that exceeded this

scope have been deleted: the ODE-based resource-distribution formulation, the "optimal system performance" language, the physical-versus-virtual healing-capacity discussion, and the forward-looking claims concerning digital twins and AI decision layers. A closing statement now specifies that the proposed indicators are diagnostic inputs to subsequent decision analysis and do not by themselves prescribe interventions or establish causal policy effects. The exploratory state-level maps are likewise reinterpreted with an explicit warning against reading them as resilience rankings. See Sections 5.2 and 6.

[R2 - 3] There is also a need for a clearer and more structured presentation of the case studies and datasets used for validation. The paper introduces three different datasets, but they appear in different sections without sufficient explanation or continuity. For instance, Figure 3 is presented on page 8, but referenced only on page 7 and 14 in the manuscript. The explanation provided on page 7 is not clear; in particular, Figure 3d does not clearly demonstrate the three recovery regimes described in the text. If such regimes are indeed present, they need to be explicitly identified and thoroughly explained.

We made three changes. First, a short overview paragraph at the start of Section 4 introduces the three case studies through the Figure 1 roadmap: they are complementary applications rather than identical replications—the COVID-19 case provides the primary quantitative evaluation, while the heat and Japan cases are transfer illustrations. Second, a concise reading-guide paragraph accompanies the heat figure (Figure 4 in the revised numbering, after insertion of the roadmap as Figure 1), stating the role of each panel, the axes and fitted relation of panel (d), and how the departure from the initial approximately linear relationship is identified on the fitted curve. Third, the three regimes in panel (d) are now explicitly identified in the text: the approximately linear regime corresponds to the segment of the fit with near-constant slope, the nonlinear regime to the departure of the fit from its linear extrapolation at larger strains, and the positive-gain regime to points with $(P - \bar{P}) < 0$ at moderate stress on the lower-left segment. The revised text interprets the departure as a descriptive regime-change diagnostic, not a failure threshold. See Sections 3.1.2 and 4 and Figure 4.

[R2 - 4] The use of temperature/heat data in the recovery context is also unclear. It appears that a queueing-type model is being applied, but key elements are not well defined. What constitutes the "disruption" in this context? How are demand and capacity interpreted? How is system capacity represented using a piecewise linear approximation? These aspects are not adequately described, making it difficult to understand or assess the validity of the approach.

The reviewer's questions—what constitutes the disruption, how demand and capacity are interpreted, and how capacity is represented by a piecewise-linear approximation—exposed that the answers depended on an unobserved, modeled dissipation process with imposed boundary values and a closure equation. In the revision we removed that mechanism entirely rather than elaborate it. The heat application now uses only observed quantities: episodes are identified by the temperature-based criterion with onset t_0, end t_3, and peak-irradiance time t_2; the disruption stress is the cumulative solar loading $D = \int_{t_0}^{t_3} \lambda(\tau)\, d\tau$; and the response persistence is the post-peak duration $P = t_3 - t_2$. The stress–strain analysis then uses $\sigma = D$ and $\varepsilon = P - \bar{P}$. No piecewise dissipation function, boundary conditions, or queue-closure equation remain, the component has been retitled "Extreme-Heat Episode Stress–Strain Analysis" (Section 3.4), and the manuscript states explicitly that no time-resolved dissipation process or surface-energy balance is estimated. See Sections 3.4 and 4.1.

[R2 - 5] In addition, the definition of the yield point does not appear to be consistent across the three case studies. While the treatment of COVID-19 mobility data is relatively clear, the other applications lack sufficient detail and methodological transparency.

We agree, and the revision resolves the inconsistency by removal rather than by unification. The offset-based definition introduced in our earlier revision required a calibration tolerance δ^* that would itself demand justification and sensitivity analysis, and the Japan case never estimated a yield point, so a "unified" definition remained internally inconsistent. The revised manuscript drops the yield point as

an estimand throughout: departures from the initial approximately linear stress–strain relationship are described as descriptive regime-change diagnostics (Sections 2.1, 3.1.2, and 4.3), the yield-point column has been removed from the land-cover table (Table 3, which now reports B, E, and R^2 only), and no case claims a physical failure threshold or irreversible adaptation. Consistency across the three cases is thereby achieved: none of them estimates a yield point.

[R2 - 6] The Queue-Based Temperature Model (QTM) is particularly underdeveloped. Several equations are introduced without derivation, intuition, or implementation details, which limits both interpretability and reproducibility.

This concern is addressed jointly with Comment 4 above: the under-developed model has been removed rather than developed further. The former QTM formulation—the latent dissipation function, its piecewise-linear specification, the queue-closure condition, and the derived capacity quantities—no longer appears in the manuscript. Reproducing the heat application now requires only the temperature-based event criterion and two direct computations on observed data, D and P, both defined in Section 3.4 with their units. This removes the reproducibility burden at its source. See Section 3.4.

[R2 - 7] Figure 5c does not exist, while it is referenced in the paper.

We corrected the inconsistency by removing the reference to panel (c) and revising the caption and corresponding text to refer only to the displayed panels. In the revised numbering this figure is Figure 5. See Figure 5.

Highlights

- Bridges traffic flow capacity theory and multi-sector urban infrastructure resilience
- Quantifies recovery via single-period capacity, elasticity, yield points, and networks
- Validates on COVID-19 mobility, extreme heat, and Japan earthquake recovery data
- Shows multi-regime behavior and queues apply beyond transportation to resilience

Manuscript Click here to view linked References

From Traffic-Flow Capacity to Recovery Capacity: An Episode-Based Framework for Multi-Sector Urban Resilience

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ABSTRACT

Urban systems experience disruptions that differ in magnitude, duration, and final recovery outcome. Existing resilience measures commonly emphasize performance loss, recovery duration, or trajectory area, but they do not directly represent the peak rate at which a disrupted system recovers. Inspired by the traffic-flow view of capacity as a dynamic, state-dependent peak rate, this paper develops an episode-based recovery-capacity framework. A Double Quadratic Queue (DQQ) procedure is used as an episode filter to identify critical transition points, and a Linear-Exponential Adaptation (LEA) model represents the recovery trajectory. Single-period recovery capacity is defined as the maximum healing rate, $R_{cap} = abe^{-c}$, while multi-episode stress–strain relationships provide complementary descriptive measures of disruption sensitivity and transition loss or gain. Model comparison using Google COVID-19 Community Mobility Reports shows that LEA generally provides lower fitting and prediction errors than benchmark recovery models. Extreme-heat episodes in Phoenix and post-earthquake infrastructure restoration in Japan are included as transfer illustrations with case-specific variables and interpretations. The resulting framework is descriptive and diagnostic rather than causal, and it provides a focused connection between observed recovery episodes and dynamic capacity analysis.

In Memory of Professor Hani S. Mahmassani (1956–2025)
Whose pioneering work in dynamic traffic assignment and network modeling continues to illuminate the science of complex urban systems

1. Introduction and Research Motivation: Linking Urban Resilience and Dynamic Capacity Modeling

Transportation resilience research provides many measures of performance loss, recovery duration, robustness, and trajectory area, but observed disruption episodes are rarely tied to an explicit representation of recovery *capacity*. This paper focuses on that missing connection by linking episode identification, recovery trajectories, and peak healing rates within a common capacity-oriented framework.

Recovery processes, marked by a gradual return to pre-disruption levels, are influenced by resilience (Holling, 1973), rapidity (Cimellaro et al., 2010), robustness (Doyle and Carlson, 2002), and adaptability (Holling, 1979). Understanding these dynamics is significant in shaping recovery outcomes, yet their connection to recovery process analysis has historically been limited. While Holling's resilience focuses on adaptability and persistence, engineering resilience models (e.g., trapezoids and triangles) simplify system recovery into quantifiable metrics (Holling, 1996). However, the complexity of socio-environmental resilience (Gu et al., 2020; Li et al., 2015; Nieves-Melendez and

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De La Garza, 2016), especially in urban systems facing climate change, calls for integrated, adaptive approaches that capture both human behavior (Brown et al., 2022) and environmental interactions (Navarro et al., 2016; Markolf et al., 2019; Li et al., 2019). Foundational studies by Mahmassani et al. (1991) and Mahmassani (2001) established dynamic traffic assignment and simulation as integral tools for understanding time-dependent network behavior, emphasizing how information, traveler decisions, and system management interact under nonstationary conditions.

Recent work has focused on capacity quantification using resilience triangular areas (Hong et al., 2021) and integrated resilience metrics (Francis and Bekera, 2014; Dong et al., 2023), examining new equilibrium points and the ratio between system performance and transition loss in disrupted states (Henry and Ramirez-Marquez, 2012). Mahmassani and Peeta (1993) demonstrated how adaptive information and control can accelerate network recovery under dynamic user-equilibrium and system-optimal conditions—an idea that resonates with our recovery-capacity formulation. However, many studies using V-shaped recovery models for single-period transitions overlook the nuanced relationship between short-term responses and broader disruption recovery curves, particularly when disruptions are repeated or involve properties shared across similar communities.

Global frameworks operationalize these concepts at scale. The Sendai Framework for Disaster Risk Reduction (United Nations Office for Disaster Risk Reduction, 2024a,b; United Nations General Assembly, 2016) emphasizes capacity-building in urban systems, where capacity encompasses the strengths, resources, and knowledge enabling disaster risk reduction (United Nations General Assembly, 2016). By March 2024, 160 countries had reported progress, with 80 completing full assessments. Urban systems operate within finite capacities determined by infrastructure, resources, and population demands. System carrying capacity can be modeled using approaches like the delayed generalized logistic model (Suweis et al., 2015) and point process analysis (Ito et al., 2024), providing insights into recovery dynamics that shape disaster risk reduction strategies.

Resilience in urban systems can be conceptually compared to stress-strain relationships in material science. Just as **Young's modulus** and **Hooke's law** describe how materials respond to stress and return to their original state after deformation, urban systems experience "stress" during disruptions and possess elastic recovery capacity before requiring adaptive changes. We adopt this analogy as a phenomenological and organizing device—not as a claim that urban recovery obeys the same conservation laws as continuum mechanics or traffic flow. Traffic flow theory rests on well-defined physical conservation laws (Lighthill and Whitham, 1955; Daganzo, 1994), whereas socio-economic recovery is governed by heterogeneous agents, institutions, and path dependencies. Accordingly, we treat the traffic-flow and material-mechanics parallels primarily as a source of structural inspiration, while empirical calibration and validation rely on observational recovery data. The connection to Mahmassani's work on commuter decision dynamics (Mahmassani and Herman, 1990) and connected autonomous vehicles (Talebpour and Mahmassani, 2016) is therefore made in this softer sense: shared regularities of multi-regime behavior, demand-supply feedback, and behavioral adaptation, rather than transfer of physical laws.

Research Gaps. Despite extensive progress in transportation and infrastructure resilience research, two gaps remain.

Gap 1 — Underdeveloped capacity interpretation of recovery. Existing resilience measures emphasize disruption magnitude, robustness, recovery duration, or trajectory area, but less attention has been given to recovery as a dynamic capacity process characterized by a peak healing rate and a state-dependent trajectory.

Gap 2 — Missing bridge between episode trajectories and cross-episode response. Individual recovery models describe how one system evolves after disruption, whereas comparative resilience analyses examine differences across episodes, sectors, and regions. A common representation is needed to connect episode-level recovery pace with transition loss or gain and multi-episode stress–strain patterns.

This paper addresses these gaps through a focused episode-based framework. DQQ is used as a preprocessing filter to identify critical transition points; LEA represents the recovery trajectory and its peak healing rate; and stress–strain regressions provide a complementary descriptive comparison across episodes. The framework is evaluated primarily with mobility data and illustrated using extreme-heat and post-earthquake infrastructure recovery trajectories.

Figure 1 provides the organizing roadmap for the manuscript. The columns represent the progression from observations and datasets to episode encoding, recovery-capacity modeling, and decision-oriented interpretation, while the rows distinguish the observation, analytical, and application layers. The three case studies occupy complementary portions of this structure. The central analytical contribution is the recovery-capacity layer connecting observed disruption episodes to interpretable trajectory and cross-episode measures. This layered, forward-passing organization—each stage passing structured quantities to the next—follows the flow-through computational-graph view of multi-layer transportation network models (Zhou et al., 2025), here used only as an organizing device rather than an implemented optimization graph.

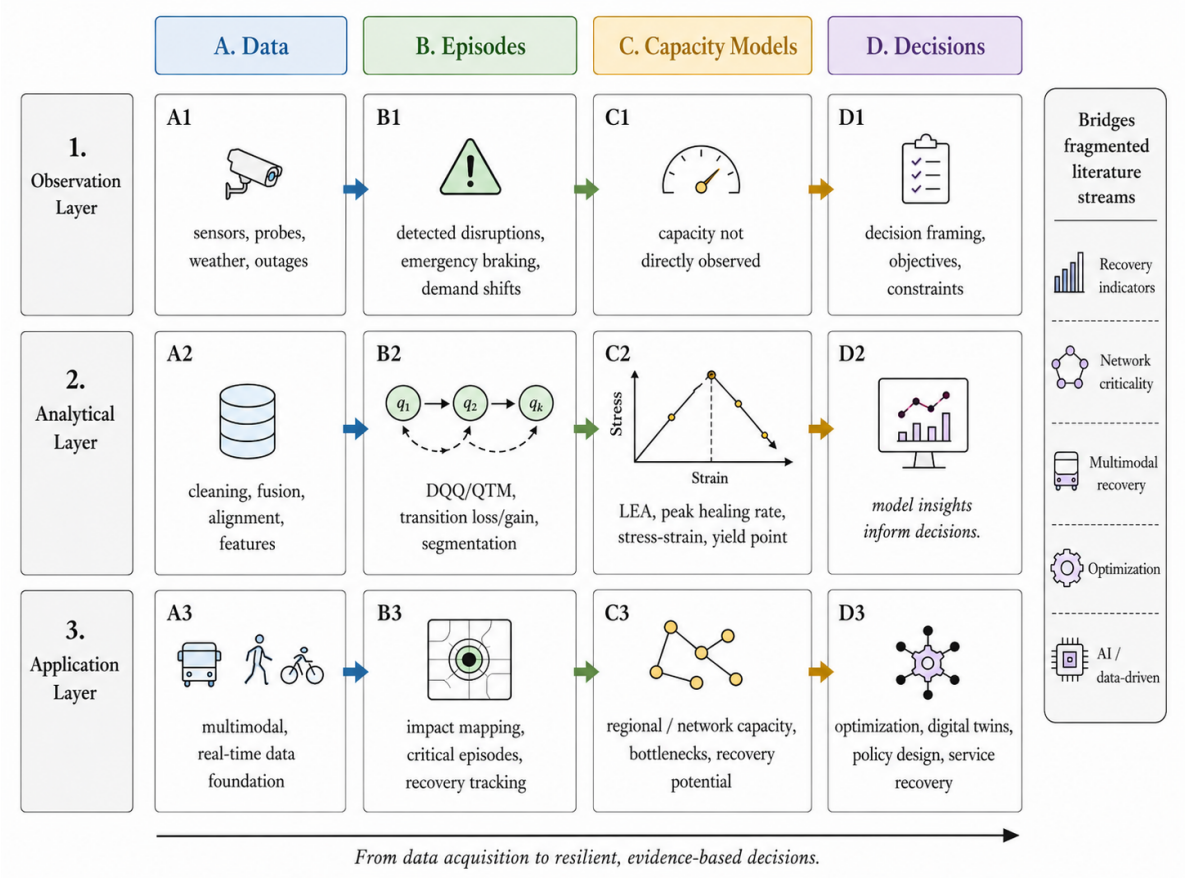


Figure 1: Roadmap from disruption data to resilient decisions. Columns represent four functional stages—data, episodes, capacity models, and decisions—and rows represent three levels of abstraction—observation, analytical formulation, and application. DQQ-based episode filtering converts raw disruption observations into structured recovery episodes; the capacity layer uses LEA peak healing rates and descriptive stress–strain relationships; and the application layer supports comparative interpretation across the three case studies.

To translate the roadmap into an analytical formulation, the paper distinguishes two complementary levels of recovery-capacity analysis:

1. Single-Period Recovery Capacity (R_{cap}): The maximum healing rate within an individual disruption–recovery episode, summarizing the intensity and timing of recovery.

2. Multi-Episode Elastic Response (E and B): A descriptive stress–strain relationship across multiple episodes, where E is the fitted slope magnitude and $B = H/E$ is the strain-axis intercept. These quantities are case-specific comparative indicators rather than universal physical constants.

The remainder of this paper is organized as follows. Section 2 establishes the conceptual foundation by mapping dynamic traffic-flow and network-modeling principles to multi-sector recovery. Section 3 presents the mathematical formulation of the LEA model, the stress-strain framework, and the extreme-heat episode stress–strain analysis. Section 4 describes data sources, preprocessing, and calibration results. Section 5 reports model comparison and applications across mobility systems, thermal resilience, and post-earthquake infrastructure recovery. Section 6 discusses policy implications, and Section 7 concludes with future research directions.

2. Conceptual Foundation: From Dynamic Traffic Capacity to Multi-Sector Recovery Capacity

Our capacity-oriented framework extends Professor Hani S. Mahmassani's foundational insights on traffic flow capacity as a *dynamic* property of complex systems. His work demonstrated that capacity emerges from interactions of demand, supply, behavior, and information, with systems evolving through distinct regimes (Mahmassani and Herman, 1984; Mahmassani et al., 1984; Mahmassani and Peeta, 1987; Mahmassani, 2001). We generalize these principles from traffic networks to multi-sector urban recovery through two core concepts: (i) capacity as a state-dependent peak rate on a dynamic trajectory; and (ii) recovery as a demand–supply adjustment process shaped by behavioral learning and system context.

Figure 2 illustrates the parallel structure connecting these traffic flow concepts to our recovery capacity framework.

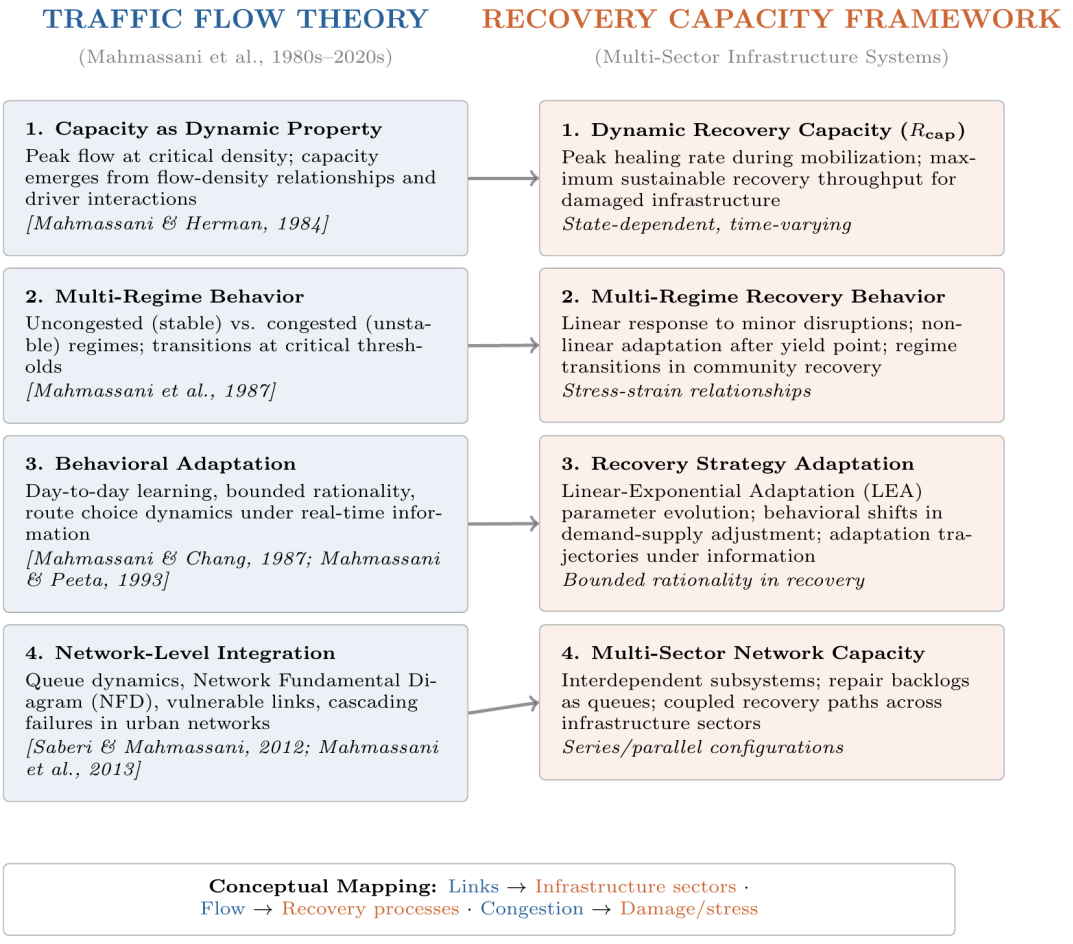


Figure 2: Conceptual bridges between traffic-flow theory and the recovery-capacity framework. Selected principles from Mahmassani's traffic-flow research—dynamic capacity, multi-regime behavior, behavioral adaptation, and state dependence—are mapped to corresponding elements of episode-based recovery analysis. The arrows indicate conceptual parallels rather than universal physical equivalence.

2.1. Capacity as Dynamic, State-Dependent Peak Performance

Building on the Herman-Mahmassani tradition (Herman and Prigogine, 1979; Mahmassani and Herman, 1984), Mahmassani's investigations demonstrated that capacity is not a fixed link attribute but an emergent outcome of flow-density-speed relationships and driver interactions. The fundamental diagram embodies this structure: an uncongested

regime with increasing throughput, a critical range where capacity is attained, and a congested regime where additional demand reduces performance (Mahmassani, 2005; Saberi and Mahmassani, 2012). These diagrams serve as compact analytical representations linking microscopic interactions to macroscopic performance through conservation laws and interaction rules.

We adopt this principle for recovery systems: capacity represents the *peak rate on a dynamic trajectory*, tied to specific operating regimes and shaped jointly by demand, supply, and state-dependent response. In traffic, this peak corresponds to maximum flow at critical density. In recovery, it corresponds to the maximum healing rate at peak mobilization: $R_{cap} = h_{max}(t = b)$. Across multiple episodes, departures from an initial approximately linear stress–strain relationship are interpreted as descriptive regime changes rather than universal physical thresholds.

The stress–strain analogy provides a descriptive bridge between traffic physics and recovery dynamics. Disruption magnitude plays the role of stress, while recovery duration or net loss/gain represents strain. Different sectors may exhibit distinct fitted relationships, much as corridors exhibit distinct macroscopic fundamental diagrams (Saberi et al., 2014; Mahmassani et al., 2013). The purpose is to identify empirical regularities in response sensitivity, not to claim physical equivalence or irreversible material change.

Having established capacity as a dynamic, state-dependent property emerging from demand-supply interactions at critical operating points, we now examine how systems evolve toward new equilibria through behavioral adaptation—a process central to both traffic recovery and multi-sector resilience.

2.2. Recovery as Behavioral Adjustment and Demand-Supply Co-Evolution

This research tradition extends beyond link-level physics to system dynamics of demand-supply co-evolution. Day-to-day models treat network evolution as behavioral adjustment under changing conditions (Mahmassani and Chang, 1987; Mahmassani and Herman, 1990). Travelers experiment, learn, and adapt rather than solving one-shot optimization problems. Capacity, travel times, and route choices co-evolve as demand (route and departure choices) responds to supply (infrastructure, control, congestion). Disruptions can shift underlying attractors, producing new equilibria (Hu and Mahmassani, 1997; Mahmassani and Liu, 1999; Zhou and Mahmassani, 2007).

This integrated view naturally extends to multi-sector recovery. Recovery trajectories encode how populations, firms, and institutions re-balance demand and supply: which activities restart, which are substituted, and which are permanently transformed. Effective recovery capacity thus reflects both physical and behavioral limits—how quickly communities change routines, reconfigure supply chains, and adopt new technologies. Regions that adjust rapidly resemble networks where travelers quickly re-route; regions that remain stuck resemble inefficient, overloaded equilibria. Route choice learning and recovery strategy adaptation share common features: bounded rationality, feedback, and gradual convergence toward new balances.

Different recovery regimes—returning to baseline, stabilizing below it, or stabilizing above it—parallel shifted traffic equilibria under new information or constraints. Advanced Traffic Information Systems (ATIS) improve routing and system performance by providing real-time conditions (Mahmassani and Peeta, 1993); similarly, real-time capacity assessment enables adaptive recovery resource allocation.

Table 1 systematically maps these core concepts from Mahmassani's framework to our recovery capacity approach, demonstrating structural parallels between transportation network performance and multi-sector resilience.

2.3. Implications for Multi-Sector Recovery Research

The dynamic-capacity and demand–supply perspectives developed in traffic-flow research provide useful structural parallels for analyzing multi-sector recovery, while their physical interpretation remains case-dependent. Their use outside transportation requires explicit attention to heterogeneous agents, institutional change, path dependence, and sector-specific dimensions, as discussed next.

2.4. Boundary Conditions and Scope of the Theoretical Mapping

The mapping from traffic-flow concepts to multi-sector recovery is intended as a phenomenological and organizing analogy rather than a transfer of physical laws. Traffic flow is governed by vehicle-conservation relationships (Lighthill and Whitham, 1955; Daganzo, 1994), whereas socio-economic recovery involves heterogeneous households, firms, institutions, substitutions, and external interventions. In recovery applications, an inflow–outflow representation should therefore be interpreted as an accounting relationship defined for a particular sector and observation period, not as a universal conservation law.

Table 1
Methodological bridges connecting traffic flow theory to recovery capacity analysis. The mappings indicate structural similarities and case-specific modeling transfers rather than universal physical equivalence.

Concept	Traffic Flow and Networks	Urban Recovery (This Work)	Nature of Relationship
Dynamic Capacity	Maximum flow at critical density: $q_{max} = f(k_{critical})$	Maximum healing rate at peak mobilization: $R_{cap} = h_{max}(t = b)$	Structural analogy
Multi-Regime Behavior	Uncongested vs. congested; transition at critical density	Approximately linear vs. nonlinear response across episodes	Shared phenomenology
Behavioral Adaptation and Demand-Supply Co-Evolution	Route and departure choice learning with inertia and bounded rationality; day-to-day equilibrium	Recovery trajectory intensity, timing, and shape summarized by LEA parameters a, b, and c; equilibrium represented through transition loss or gain	Conceptual correspondence
State and Path Dependence	Capacity varies with flow, weather, incidents, and behavior	Recovery trajectory conditioned on disruption magnitude, resources, and history	Conceptual correspondence

The framework is most appropriate when a meaningful baseline, a bounded disruption episode, and a measurable recovery trajectory can be identified. Parameters such as E, H, and B should be interpreted as case-specific descriptive quantities rather than transferable physical constants. Cross-case comparisons should rely on normalized indicators or qualitative structural comparisons, while causal mechanisms and institutional changes require separate analysis.

3. Modeling Framework and Mathematical Formulation

This section presents the mathematical formulation of our capacity-oriented framework. For a comprehensive reference of all mathematical notation and symbols used throughout this paper, readers may consult Table A1 in the Appendix.

3.1. Model Structure and Key Equations

3.1.1. Recovery Trajectory, Healing Rate Function, and Single-Period Recovery Capacity Analysis

Analyzing system recovery requires a framework that accounts for non-linear dynamics and **transition losses and gains**. Real-world systems often fall short of pre-disruption states (transition loss) or surpass them (transition gain), achieving higher operational efficiency. We propose a **healing rate-based framework** building on hazard models to capture these dynamics, calculating both the **Instantaneous Healing Rate** ($h(t)$) and overall **Recovery Capacity**.

We demonstrate the framework using the *Google COVID-19 Community Mobility Reports* dataset tracking movement trends during the pandemic. Figure 3a shows percent changes in U.S. activity levels (2/15/2020–12/31/2020). Using the DQQ model (Tang et al., 2025) to identify boundary equilibrium and stationary points, we delineate three transition states for disrupted systems (Figure 3b):

1. **Normal**: The system returns to its pre-perturbation baseline.
2. **Transition Loss**: The system stabilizes below baseline, indicating a permanent shift or "new normal."
3. **Transition Gain**: The system recovers beyond baseline, establishing a new, improved norm.

The healing-rate model focuses on three core concepts: *Instantaneous Healing Rate* $h(t)$, *Recovery Speed Function* $f(t)$, and *Recovery/Resilience Trajectory Area* RTA (Haimes, 2009; Hosseini et al., 2016). Here, $h(t)$ constitutes the essence of the LEA model, incorporating both linear and exponential components. Dependent on Sub-Remaining Population $S(t)$, $h(t)$ provides an immediate measure of recovery likelihood, reflecting conditional probability. In contrast, $f(t)$ maps the potential recovery trajectory over time. The RTA, estimated by the area enclosed by $Q_R(t)$ and line $y = \Delta$ (shaded region in Figure 3b), serves as a resilience measure (Bruneau et al., 2003; Gu et al., 2020), analogous to the V-shaped triangle area in resilience studies (Gu et al., 2020).

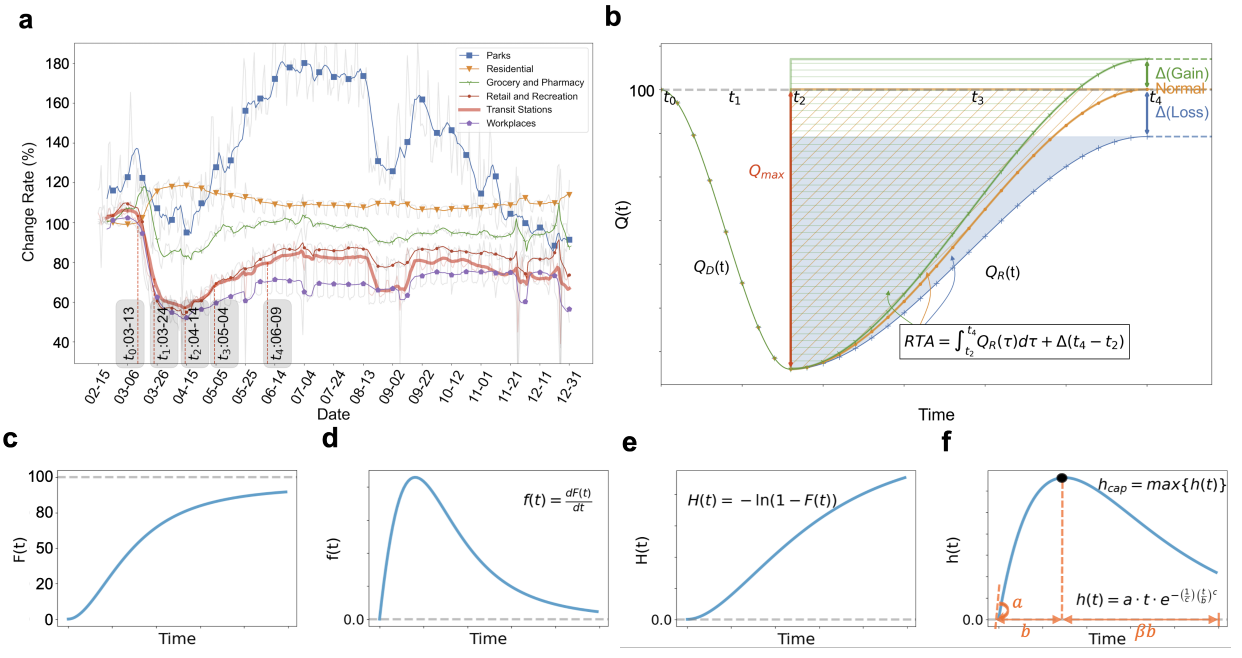


Figure 3: Data processing and methodology in LEA model | a Mobility pattern changes across six mobility types in the United States (50 states and Washington DC) in 2020. **b** Three scenarios of disruption and recovery processes, adapted from Tang et al. (2025). Mobility pattern changes $Q(t)$ include disruption $Q_D(t)$ and recovery $Q_R(t)$ processes. Boundary conditions are defined by critical steady-state time points (t_0 and t_4) where trajectory gradient ($Q(t)$) is significantly small. Points t_1 and t_3 mark maximum gradient, and t_2 corresponds to the peak/valley Q_{max}. **c–f** illustrate LEA model components in the recovery phase: **c** Forward Trajectory Function $F(t)$ normalized from $Q_R(t)$ (0–100%); **d** Sub-Remaining Function $S(t)$, indicating unrecovered system proportion (Wainer and Velleman, 2001); **e** Speed Profile $f(t)$ quantifying recovery velocity; **f** Healing Function $h(t)$ as the ratio of $f(t)$ to remaining affected population.

We formalize this through the **Linear-Exponential Adaptation (LEA) model**:

$$h(t) = a \cdot t \cdot e^{-c \cdot \left(\frac{t}{b}\right)^{\frac{1}{c}}} \tag{1}$$

where parameters define recovery phases:

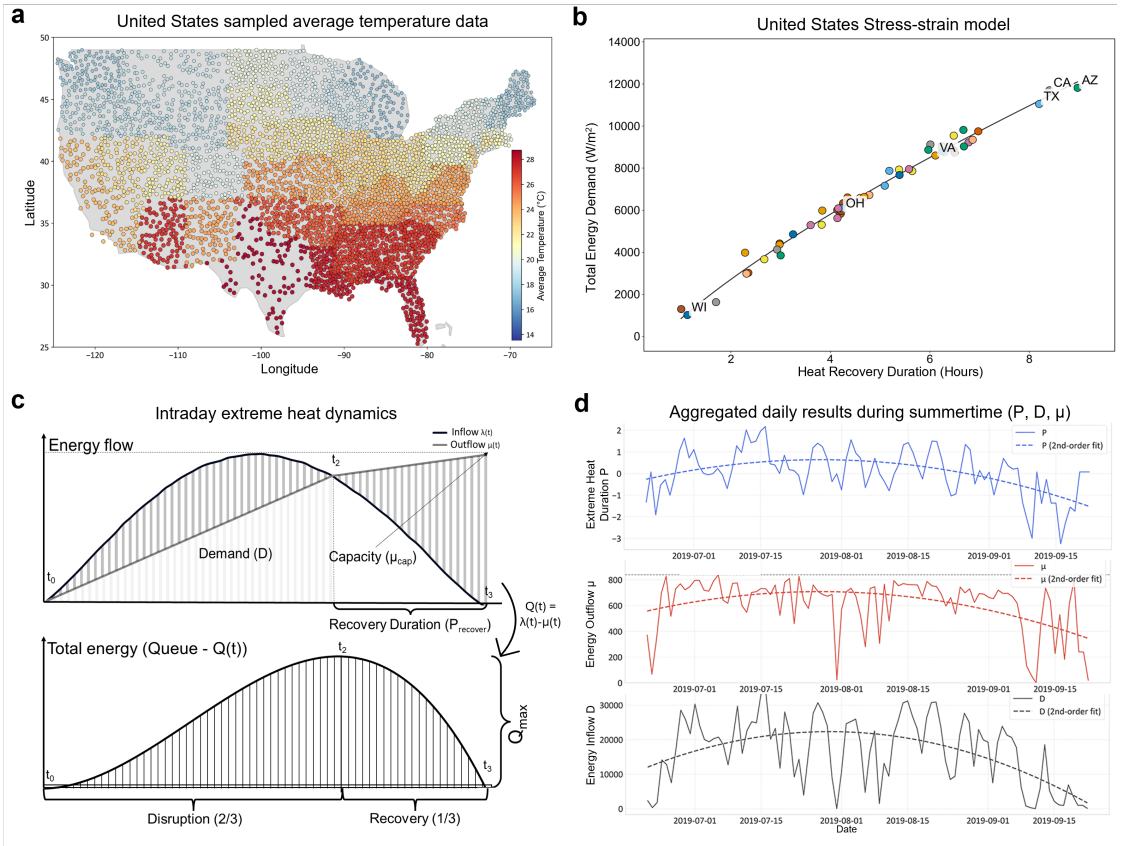
- a: Initial recovery speed.
- b: Duration of recovery exploration phase (time of peak).
- c: System's transition ability to steady state (decay factor).

The three LEA parameters admit distinct interpretations: a sets the initial mobilization intensity, since $h(t)/t \to a$ as $t \to 0^+$; b is the time of peak healing, since $dh/dt = 0$ at $t^* = b$; and c is a dimensionless decay exponent controlling how sharply $h(t)$ falls away from its peak. The three parameters thus separate the *intensity*, *timing*, and *shape* of the healing process, which two-parameter forms such as the exponential or Weibull conflate. Model comparison is also not based on in-sample SSE alone: we report out-of-sample (70%/30% split) prediction errors in Section 4.2, aggregated across 50 states × 6 mobility types, so that overfitting on any single trajectory is unlikely to explain the consistent gains.

In the **steady state** phase, the system stabilizes as healing rate $h(t)$ approaches zero (Hosseini et al., 2016; Liu et al., 2022), reaching either full recovery or a new norm. Remaining transition gains indicate positive adaptation; their absence may signal "feedback loss" hindering full recovery.

As illustrated in Figure 3f, the **Single-Period Recovery Capacity** (R_{cap}) is the peak healing rate within the episode. Parameter b identifies the time of this peak, while a and c determine its magnitude and decay shape.

3.1.2. Using Stress-Strain Models to Understand Adaptive Capacity in Episode-Based Trajectories



e

Multi-scale Geographically Weighted Regression (MGWR) land cover coefficients in Maricopa County, Arizona

Land Cover Type	Extreme Heat Duration P	Energy Outflow μ	Extreme Heat Capcity C	Energy Inflow D
Developed (Combined Medium and High Intensity)	0.76	-0.67	-0.13	0.80
Developed, Low Intensity	0.30	-0.26	0.10	0.32
Developed, High Intensity	3.70	-3.71	2.04	3.35
Forest (Combined Deciduous, Evergreen, Mixed)	-3.45	8.41	-2.06	-1.75
Barren Land	1.25	-1.26	-0.31	1.05
Open Water	0.18	-0.21	0.10	0.17

Figure 4: Resilience metrics analysis across states using extreme-heat episode indicators | a 100 sampled locations per state across 49 states to obtain gridded (2×2 km) 2-m air temperature and Global Horizontal Irradiance (GHI) for 2019. **b** State-level data showing energy demand D (stress) vs. extreme heat duration P (strain) for U.S. heat resilience. **c** Example intraday extreme-heat dynamics. The onset t_0 and end t_3 are identified using the temperature-based extreme-heat criterion, and t_2 denotes the time of maximum GHI within the event. The quantities D and P represent cumulative radiative loading and post-peak duration, respectively. **d** Extreme heat indicators from **c** plotted over 93 summer days with second-order polynomial fit showing general trend. **e** Framework application to Maricopa County using multi-scale geographically weighted regression (MGWR) to calculate spatial correlations between land cover types and model outputs.

The stress–strain model provides a descriptive comparison of system responses across episodes. For small-to-moderate strains, the fitted relationship may be approximately linear; at larger strains, the response may depart from that initial trend. We interpret such departures as empirical regime changes rather than physical failure thresholds or evidence of irreversible adaptation.

Stress–Strain Modeling for Extreme Heat Recovery. We apply this framework to land-cover response, where stress is cumulative solar loading and strain is the relative post-peak heat duration $P - \overline{P}$, with $\overline{P}$ denoting the average duration at a location. Second-order polynomial fits are used only to describe departures from the initial approximately linear relationship.

Because Figure 4 contains both the intraday extreme-heat illustration (panel c, used later in Section 3.4) and the multi-day stress-strain regime diagnostic (panel d), we summarize its role here for clarity. Panels (a)–(c) document how the raw NLCD/NSRDB inputs are converted into episode indicators (D, P); panel (d) then plots these indicators across 93 summer days and is the diagnostic figure for the three recovery regimes; panel (e) provides the spatial MGWR companion. In particular, in Figure 4d each marker is a single summer day, the horizontal axis is the strain ($P - \bar{P}$), the vertical axis is the stress D, and the second-order polynomial fit (solid curve) is used to describe departure from the initial approximately linear regime. This is an exploratory regime-change diagnostic rather than a physical failure threshold.

The three recovery regimes identified in Figure 4d.

- **Linear Recovery Regime:** For small-to-moderate strains ($P-\bar{P}$), the fitted curve is approximately linear. Within this regime, the response is summarized by Young's modulus E, defined as the magnitude of the fitted stress–strain slope. For the linear relation $\sigma = H - E\varepsilon$,

$$E = -\frac{d\sigma}{d\varepsilon}, \tag{2}$$

  where $E > 0$ is the magnitude of the fitted slope and H is the stress-axis intercept. In Figure 4d, this regime corresponds to the portion of the fitted relationship where the slope is approximately constant.

- **Nonlinear Recovery Regime:** At larger strains, the fitted response departs from the initial approximately linear relationship. In Figure 4d, this regime appears as the departure of the polynomial fit from its linear extrapolation. This departure is interpreted as a descriptive nonlinear-response threshold and does not, by itself, establish irreversible structural change or permanent adaptation.

- **Positive Gain Regime:** For certain land-cover types and in certain mobility sectors, the strain measured against the average recovery duration can be *shorter* than baseline, producing points with $(P-\bar{P}) < 0$ at moderate stress and indicating enhanced post-disruption performance rather than loss. In Figure 4d these correspond to points on the lower-left segment of the fit; the analogous mobility case appears in Figure 6b for Residential activity.

The fitted indicators vary across land-cover types. Table 3 reports the within-case quantities E, B, and R^2. These values are descriptive summaries of the fitted relationships and should not be interpreted as universal material properties or causal measures of adaptation.

The units of E, H, and B depend on the definitions of stress and strain. In the mobility analysis, $\sigma = |Q_{\max}|$ and $\varepsilon = \Delta$ are relative deviations, so E, H, and B are dimensionless. In the thermal analysis, E has units of W/m^2, H has units of W·h/m^2, and B has units of hours. The Japan case does not estimate E.

For the LEA model, $R_{cap} = abe^{-c}$ has units of inverse time. Therefore, these quantities are interpreted within each application; cross-case comparisons emphasize structural patterns rather than direct numerical equivalence.

3.2. Recovery trajectory represented by Linear-Exponential Adaptation model

The fraction of the population still disrupted at time t is $S(t) = e^{-H(t)}$. Here, the cumulative healing (recovery) capacity $H(t)$ is derived from the integral of the instantaneous healing rate $h(t)$ from an initial time t_2 to t. The recovery fraction is expressed as $F(t) = 1 - S(t)$. In the realm of recovery dynamics, the recovery speed function, denoted as $f(t)$, represents the likelihood of system recovery at an exact time t, given by $h(t) = \frac{f(t)}{S(t)}$. To avoid division by zero or enormous values of $h(t)$ when $S(t)$ approaches zero, a regularization approach of $S(t) = \max\{\epsilon, S(t)\}$ is utilized, where ϵ is a small positive value which can be determined under deviation statistics. Given $F(t)$'s range, normalization is executed by $\frac{Q_R(t)}{M}$, where $M = \max\{|Q_{max}|, |Q_{max}| + \Delta\}$. In other words, in the context of system loss, M is the absolute value $|Q_{max}|$, and for normal conditions and system gain, M is defined as $|Q_{max}| + \Delta$, where Δ indicating the system-induced demand. The model is summarized by the compact chain:

$$H(t) = \int_0^t h(\tau)\, d\tau, \qquad S(t) = e^{-H(t)}, \qquad F(t) = 1 - S(t), \qquad f(t) = h(t) \cdot S(t). \tag{3}$$

Property 1: Within the LEA specification, the single-period recovery capacity is determined by the fitted parameters a, b, and c. The empirical associations of R_{cap} with disruption magnitude $|Q_{max}|$ and transition loss or gain Δ are examined separately in the multi-episode analysis.

Recovery capacity R_{cap} emerges as a critical indicator of a system's resilience. It can be expressed as the maximum healing rate at $t = b$.

$$R_{cap} = a \cdot b \cdot e^{-c} \tag{4}$$

It exhibits significant variation across states and mobility types, thus reflecting diverse recovery dynamics. Regions with higher healing capacities are *empirically associated* with faster observed return to baseline in the data analyzed here; we do not claim a causal mechanism, since these patterns are estimated from observational mobility records and may be confounded by unobserved policy, demographic, or epidemiological factors.

In practical terms, recovery capacity can inform real-world transportation recovery metrics, such as the time required for transit ridership to bounce back post-disruption, or the adaptability of bus schedules to new norms. As depicted in Figure 3f, recovery capacity measurements can be correlated with cumulative recovery trajectories, represented in terms of $F(t)$ and $S(t)$.

3.3. Peak Recovery Capacity

Derived forms of LEA model to analyze recovery trajectory. Here, the focus is on the recovery time frame (t_2 to t_4) from the lowest point to the steady state. In contrast to traditional hazard models, the LEA model is able to capture the transition loss and gain in derived forms of the trajectory. As shown in Figure 3, the transition loss case exhibits a considerably slower healing rate compared to both the normal and gain cases.

The first order derivative of $h(t)$ with respect to t is:

$$\frac{dh}{dt} = -ae^{-c\left(\frac{t}{b}\right)^{\frac{1}{c}}}\left(\left(\frac{t}{b}\right)^{\frac{1}{c}} - 1\right) \tag{5}$$

Setting $\frac{dh}{dt} = 0$ and solving for t, we find that the maximum healing rate as the capacity occurs at $t = b$.

3.4. Extreme-Heat Episode Stress–Strain Analysis

Extreme-heat episode representation. We identify bounded extreme-heat episodes using the observed temperature criterion, with event onset t_0 and event end t_3, and with t_2 denoting the time of maximum Global Horizontal Irradiance (GHI) $\lambda(\tau)$ within the event. For each episode, cumulative solar loading

$$D = \int_{t_0}^{t_3} \lambda(\tau)\, d\tau \tag{6}$$

represents disruption stress, while post-peak heat duration

$$P = t_3 - t_2 \tag{7}$$

represents the persistence of the thermal response. The analysis does not attempt to estimate a time-resolved physical dissipation process or a complete surface-energy balance. Instead, it uses the observable pair (D, P) to examine whether larger environmental loading is associated with longer recovery duration and nonlinear regime changes. In the multi-episode stress–strain analysis, we take

$$\sigma = D, \qquad \varepsilon = P - \bar{P}, \tag{8}$$

where $\bar{P}$ is the average post-peak duration at a location.

4. Data Sources, Preprocessing, and Model Implementation

This section describes the three complementary datasets employed to validate the recovery capacity framework across mobility systems, thermal resilience, and physical-infrastructure recovery. The three case studies examine complementary components of the proposed framework rather than identical implementations of a single model. The COVID-19 mobility case evaluates LEA trajectory modeling and stress–strain relationships. The extreme-heat case uses episode indicators to examine thermal recovery and land-cover response. The Japan post-earthquake case illustrates subsystem recovery trajectories.

4.1. Data and pre-processing

All data that support the key findings in this study are available within the main text and the Supplementary Information file. Additional raw data are available from the corresponding authors upon request. Source data are provided with this paper. The mobility analysis uses the 50 U.S. states and the District of Columbia (51 spatial units), each with six mobility categories, over 2020. The state-level model-comparison and prediction results are reported for the same 51 units.

The baseline is calculated as the median value from the 5-week period between Jan 3 and Feb 6, 2020 (Google Mobility Report, 2020), with each corresponding daily value measuring the percentage deviation from that baseline median. In this figure, the lighter lines represent raw, unfiltered data while heavier lines depict the smoothed 7-day moving average data.

For the smoothed 7-day moving averages depicted in Figure 3a, this study employs a second-order polynomial approximation (Zhou et al., 2022) through the Double Quadratic Queue model (Tang et al., 2025). This model is particularly effective for accurately approximating nonlinear time series data within the specified framework (Saberi et al., 2020). While it may be possible to use alternative smoothing methods, it is important to emphasize that they must align with the established boundary conditions and the dynamics of recovery and disruption (Cimellaro et al., 2010; Faturechi and Miller-Hooks, 2014).

In analyzing the Google Mobility dataset using the LEA model, this study specifically focuses on the recovery phase following the initial wave of the pandemic in the United States. It thus tracks the period from the lowest point of mobility disruption to the eventual attainment of a steady state. Additionally, it is important to note that for privacy, Google displays percentage changes in mobility relative to the day-of-week-specific median over the baseline period across different categories of places, including retail and recreation, groceries and pharmacies, parks, transit stations, workplaces, and residential areas. The data are further protected by anonymization and differential privacy techniques, which means that exact population or venue-specific data is not available beyond relative changes in mobility across these locations. Nonetheless, the dataset provides a robust framework for examining the dynamics of mobility trends and recovery patterns during this critical period.

Relative-baseline data and the robustness of R_{cap}. The Google COVID-19 Community Mobility Reports measure relative activity changes against a fixed five-week baseline (Jan 3–Feb 6, 2020) rather than absolute activity counts. Writing $Q(t) = N(t)/\tilde{N} - 1$ for baseline level $\tilde{N}$, the proportional healing rate satisfies

$$h_{\mathrm{rel}}(t) = \frac{1}{1+Q(t)}\frac{dQ}{dt} = \frac{1}{N(t)}\frac{dN}{dt} = h_{\mathrm{abs}}(t), \tag{9}$$

i.e., under a fixed multiplicative baseline, proportional rates are invariant to rescaling, so the COVID-19 application is interpreted as *relative* recovery rather than absolute demand recovery. A non-stationary baseline may nevertheless affect the estimates, and absolute quantities such as passengers or trips would require an external count-based reference; the Japan service-availability case provides the complementary test on absolute percentages.

4.2. Model Comparison Framework and Calibration Results

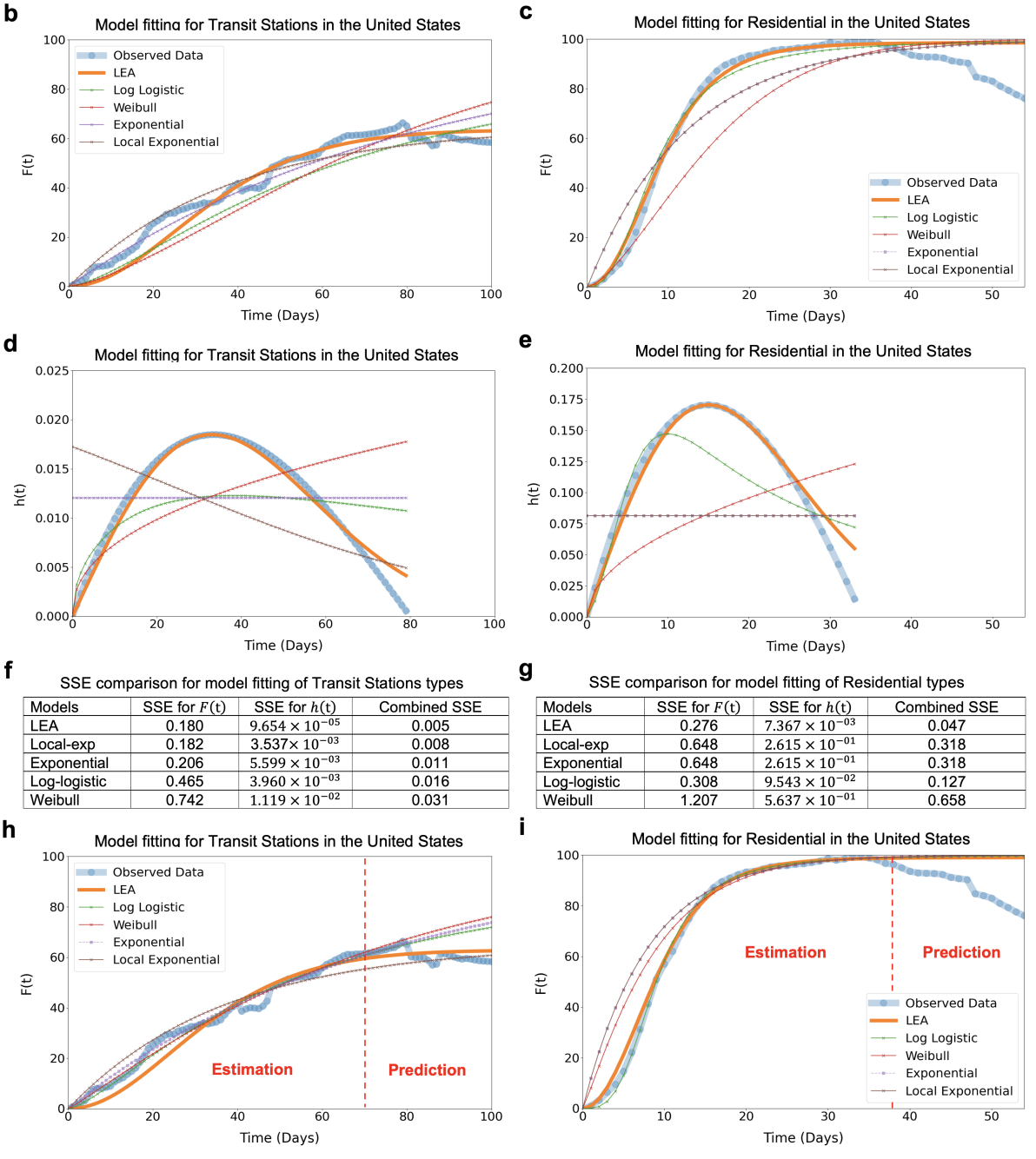
Need for Recovery Capacity Quantification: Quantifying recovery capacity, R_{cap}, is critical for understanding system resilience but is not directly observable. Thus, we require a strong underlying functional form to perform this quantification accurately. The LEA model's structure, which incorporates both linear and exponential terms, is designed to capture the peak healing rate and transition losses observed in real-world data. We analyze different hazard functions and evaluate their performance during both estimation and prediction phases.

Figure 5**a** provides a summary of the functional forms for the Forward Trajectory Function F(t) and the Healing Function h(t) for each model. The LEA model integrates both linear and exponential components, capturing complex recovery behaviors over time. The comparison across the four models allows us to explore how well each model aligns with observed mobility data.

Transition Gains and Losses: In Figure 6, we examine the distribution of transition gains and losses Δ for the first disruption and recovery period across different mobility types. Residential mobility shows a positive Δ, indicating transition gains where the system recovers beyond its pre-disruption state. Conversely, workplace mobility presents negative Δ, signifying a transition loss where recovery remains incomplete. These findings provide valuable insights into how different systems adapt to disruptions.

a

Models	Exponential	Log-Logistic	Weibull	LEA	Local Exponential
$F(t)$	$F(t) = 1 - e^{-at}$	$F(t) = \frac{t^b}{a^b + t^b}$	$F(t) = 1 - e^{-\left(\frac{t}{a}\right)^b}$	$F(t) = 1 - e^{-a\int_0^t \tau * e^{-\left(\frac{1}{c}\right)\left(\frac{\tau}{b}\right)^c} d\tau}$	$F(t) = 1 - \delta - (\sigma - \delta)e^{-at}$
$h(t)$	$h(t) = a$	$h(t) = \frac{\frac{b}{a}\left(\frac{t}{a}\right)^{b-1}}{1 + \left(\frac{t}{a}\right)^b}$	$h(t) = \frac{b}{a}\left(\frac{t}{a}\right)^{b-1}$	$h(t) = ate^{-\left(\frac{1}{c}\right)\left(\frac{t}{b}\right)^c}$	$h(t) = \frac{a(\sigma - \delta)e^{-at}}{\delta + (\sigma - \delta)e^{-at}}$



f SSE comparison for model fitting of Transit Stations types

Models	SSE for $F(t)$	SSE for $h(t)$	Combined SSE
LEA	0.180	9.654×10^{-05}	0.005
Local-exp	0.182	3.537×10^{-03}	0.008
Exponential	0.206	5.599×10^{-03}	0.011
Log-logistic	0.465	3.960×10^{-03}	0.016
Weibull	0.742	1.119×10^{-02}	0.031

g SSE comparison for model fitting of Residential types

Models	SSE for $F(t)$	SSE for $h(t)$	Combined SSE
LEA	0.276	7.367×10^{-03}	0.047
Local-exp	0.648	2.615×10^{-01}	0.318
Exponential	0.648	2.615×10^{-01}	0.318
Log-logistic	0.308	9.543×10^{-02}	0.127
Weibull	1.207	5.637×10^{-01}	0.658

Figure 5: Model comparisons and corresponding fit and prediction| **a** summarizes the functional form of $F(t)$ and $h(t)$ across various commonly utilized models, including the LEA. **b**, **c**, **d**, and **e** compare models' calibrated fits of $F(t)$ and $h(t)$ for Transit Stations and Residential respectively. **f** and **g** compare the Sum of Squared Error (SSE) for these models, where combined SSE represents an integrated error index combined by SSE for $F(t)$ and $h(t)$. **h** and **i** show model prediction for last 30% data, while the former 70% data is used to estimate model parameters.

Data Analysis and Model Fitting: Figures 5**b** and 5**c** show the model fits of $F(t)$ for Transit Stations and Residential data, respectively. Given $F(t)$'s range, normalization is executed by $\frac{Q_R(t)}{M}$, where $M = \max\{|Q_{max}|, |Q_{max}| + \Delta\}$. In other words, in the context of system loss, M is the absolute value $|Q_{max}|$, and for normal conditions and system gain, M is defined as $|Q_{max}| + \Delta$, where Δ indicating the system-induced demand. In each graph, the observed data is shown alongside the fitted curves for each model. The LEA model demonstrates a closer fit across the entire trajectory, particularly during the later stages where traditional models like Exponential and Weibull show deviations from observed data. However, it is important to note that this close fit could be attributed to the specific dynamics of the data used (mobility recovery during the pandemic). Thus, while the LEA model performs well under these conditions, generalizations about its broader applicability should be made with caution.

Recovery Capacity Assessment: Figures 5**d** and 5**e** further illustrate the fit of the Healing Function $h(t)$ for Transit Stations and Residential types, respectively. Here, the recovery capacity is defined as the peak of $h(t)$, representing the system's maximum rate of recovery after disruption. In both cases (Transit Stations and Residential), the LEA model captures this peak with greater accuracy compared to the other models, which tend to exhibit larger deviations during the decline phase of healing.

Sum of Squared Error (SSE) Analysis: To quantitatively assess the models' performance, Figures 5**f** and 5**g** present the SSE for the model fitting of F(t) and h(t). The combined SSE, integrating errors for both functions, shows that the LEA model results in lower combined SSE for the given datasets. It should be noted that this does not imply universal superiority, but it suggests that the LEA model is well-suited for fitting the observed recovery data under the specific conditions evaluated.

Predictive Capability: Figures 5**h** and 5**i** illustrate the models' predictions for the last 30% of the mobility data of Transit Station and Residential in the United States, with the first 70% used for parameter estimation. While all models provide reasonable estimates, Table A2 shows that LEA attains the lowest combined fitting SSE for five of six mobility types, the lowest $F(t)$ prediction SSE for five of six types, and the lowest transition-loss prediction SSE for all six types. Table 4 provides detailed results for Transit Stations and Residential activity. Because no inferential significance test is reported, these results should be interpreted as comparative predictive performance rather than statistical significance. However, this does not imply that the LEA model will always outperform other models in every context. The context of mobility recovery during a pandemic may uniquely favor the LEA model's multi-phase recovery structure.

4.3. Multi-Episode Stress-Strain Analysis and Young's Modulus

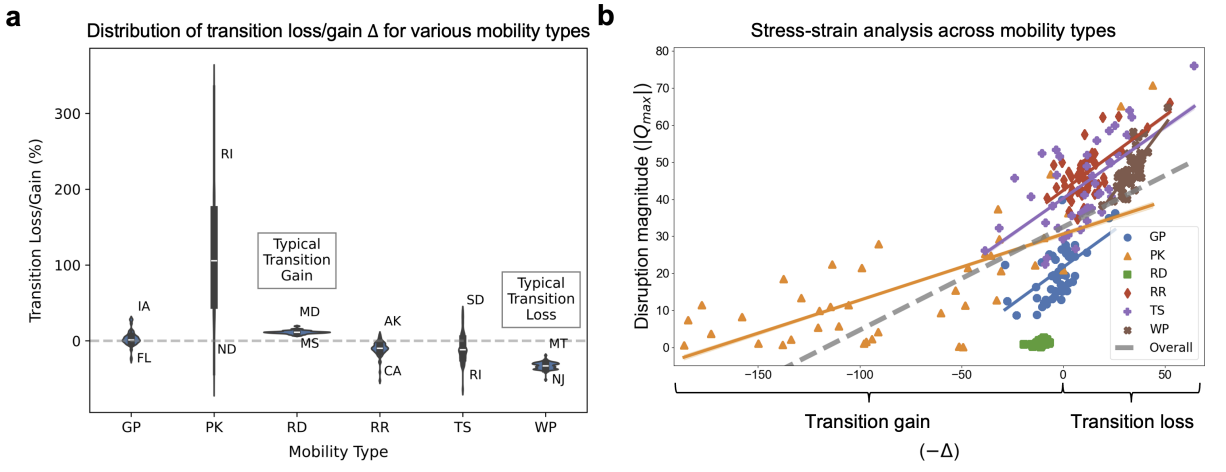


Figure 6: Stress–strain analysis across mobility types | a Distribution of transition gain or loss Δ across six mobility types: Grocery and Pharmacy (GP), Parks (PK), Residential (RD), Retail and Recreation (RR), Transit Stations (TS), and Workplaces (WP). Residential mobility generally exhibits $\Delta > 0$, corresponding to transition gain, whereas Workplace mobility generally exhibits $\Delta < 0$, corresponding to transition loss. **b** Relationship between disruption magnitude $|Q_{\max}|$ and transition gain/loss Δ. For the fitted relation $|Q_{\max}| = H - E\Delta$, E is the magnitude of the fitted slope, H is the stress-axis intercept, and $B = H/E$ is the strain-axis intercept.

This subsection explores the underlying mechanisms driving recovery patterns in urban systems, particularly the transitions between linear and nonlinear recovery regimes. We hypothesize that the relationship between disruption magnitude, transition loss or gain, and single-period recovery capacity is mobility-type specific and varies across different recovery phases.

We assess whether the relationships among disruption magnitude, transition loss or gain, and recovery capacity differ across mobility types and whether an approximately linear response can be identified over a limited range.

4.3.1. Urban Heat Resilience: Extreme-Heat Episode Land Cover Analysis

Data from the National Land Cover Database (NLCD) and the National Solar Radiation Database (NSRDB) provide gridded 2-m air temperature and Global Horizontal Irradiance over 2 km × 2 km cells. Extreme-heat events are identified using the temperature-based 97.5th-percentile criterion. Within each identified event, the observed GHI $\lambda(t)$ defines the cumulative solar loading D, and the post-peak duration P measures the persistence of the thermal response.

This episode-based approach provides resilience indicators, namely Extreme Heat Recovery Duration P and Total Inflow Demand D. Figure 4c illustrates these indicators during the summer period, offering insights into the system's resilience dynamics and elasticity in response to increased solar radiation.

Figure 4d provides an exploratory comparison of cumulative solar loading and relative post-peak heat duration, including the fitted linear sensitivity and departure from that initial trend.

4.3.2. Mobility Recovery: Type-Specific Stress-Strain Patterns

Figure 6 provides a detailed analysis of the relationship between transition gain and loss Δ, recovery capacity R_{cap} and disruption magnitude $|Q_{max}|$, with specific attention to how these metrics vary across different mobility types. The results indicate the multi-regime and mobility-type-specific relationship between recovery capacity and disruption magnitude, where the disruption magnitude is proportional to the transition gain and loss, showing a straightforward inverse relationship between recovery capacity and disruption magnitude. In addition, the results demonstrate a nonlinear recovery pattern, suggesting that larger disruptions may overwhelm the system's inherent recovery mechanisms, resulting in steeper declines in recovery capacity.

Figure 6b presents a stress-strain analysis across six mobility types. This subplot visualizes the relationship between transition gain/loss Δ and the disruption magnitude $|Q_{max}|$. It clearly shows that different mobility types exhibit distinct recovery patterns. Mobility types like Workplaces and Residential demonstrate full transition loss and transition gain phases respectively, illustrating the variability in recovery dynamics across mobility types. As the disruption magnitude increases, the system experiences a Transition Loss where recovery slows down, further reducing recovery capacity. Table 2 summarized the slope and intercept of regression curves, where the slope is defined as Young's modulus during the linear recovery regime.

Table 2
Calibration of the stress–strain model across mobility types. The fitted relationship is $|Q_{\max}| = H - E\Delta$, where E is the slope magnitude, H is the stress-axis intercept, and $B = H/E$ is the strain-axis intercept.

Mobility type	Young's Modulus E	Response threshold H	System base B	R^2
GP	0.400	21.7	54.25	0.333
PK	0.179	30.6	170.95	0.468
RD	0.088	2.1	23.86	0.170
RR	0.403	42.4	105.21	0.408
TS	0.385	40.3	104.68	0.401
WP	0.738	23.3	31.57	0.522
Overall	0.278	32.5	116.91	0.393

In a companion state-level analysis, we examined the direct relationship between recovery capacity R_{cap} and disruption magnitude $|Q_{max}|$ for the same set of mobility types. Across all mobility types, there is a clear negative relationship between recovery capacity and disruption magnitude. As disruption magnitude increases, recovery capacity tends to decrease, reflecting the challenges in achieving a rapid recovery in highly disrupted systems. This relationship is mobility-type-specific, with all mobility types showing more pronounced decreases in recovery capacity as disruption magnitude grows.

This mobility-specific analysis demonstrates that urban systems do not recover uniformly across all types of disruptions or mobility types. The degree of recovery depends heavily on the inherent properties of each system, as well as the magnitude and nature of the disruption. The nonlinear regimes of stress-strain model observed in certain mobility types reflect the complex, multi-phase processes that urban systems undergo when faced with significant disruptions.

4.4. Spatial Resilience Patterns: State-Level Triangle Base Metric

This study introduces the **stress-strain triangle base** B to quantify urban system resilience across regions and mobility types. The metric is derived from a stress-strain model that describes the relationship between transition gain/loss (Δ) and disruption magnitude ($|Q_{\max}|$):

$$|Q_{\max}| = H - E \cdot \Delta \tag{10}$$

Here, H represents the system's response threshold, and E (Young's modulus) reflects its sensitivity to disruptions. The resilience metric B is defined as the ratio of these parameters:

$$B = \frac{H}{E} \tag{11}$$

Within a given application, $B = H/E$ is the strain-axis intercept of the fitted stress–strain relationship. A larger B indicates that the fitted relation reaches zero stress at a larger strain value. Because B retains the units of strain, it is a within-case diagnostic rather than a universal dimensionless resilience score.

Figure 7 illustrates the spatial variability of H, E, and B across the United States. The patterns are presented as exploratory diagnostics only. Because the regressions have heterogeneous explanatory power, a large or small mapped value should not be interpreted as a definitive ranking of state resilience without accompanying uncertainty and fit-quality assessment.

5. Results: Multi-Scale Validation and Applications

This section presents empirical validation of the proposed recovery capacity framework across three distinct application domains: (1) human mobility systems during COVID-19 disruption, (2) urban heat resilience across diverse land cover types, and (3) post-earthquake infrastructure recovery in Japan. We begin by validating model performance through systematic comparison with alternative recovery models. We then apply the stress-strain framework to characterize extreme heat resilience across land cover types, followed by spatial analysis of state-level resilience metrics. Finally, we analyze Japan's post-earthquake recovery experience.

Table 3 summarizes the fitted parameters for different land-cover types, including System Base (B), Young's Modulus (E), and R^2.

5.1. Model Performance Comparison and Validation Results

Table 4 compares the Sum of Squared Error (SSE) for various models used to predict Transit Stations (TS) and Residential (RD) mobility types in the United States.

5.2. Exploratory State-Level Resilience Patterns

The state-level regressions and maps are retained as an exploratory analysis of spatial heterogeneity in the fitted quantities E, H, and B. The coefficient of variation of E is approximately 0.70, indicating substantial heterogeneity rather than a universal slope. Because fit quality varies considerably across states and $B = H/E$ can become large when the fitted slope is small, these maps should not be interpreted as definitive resilience rankings. Formal state comparison would require uncertainty intervals, fit-quality screening, and a hierarchical or pooled-versus-unpooled model.

5.3. Japan Post-Earthquake Infrastructure Recovery

Following the Great East Japan Earthquake in 2011 (Reconstruction Agency, 2024), the recovery of Japan's water and transportation systems was a critical focus (Global Center for Disaster Statistics, 2024).

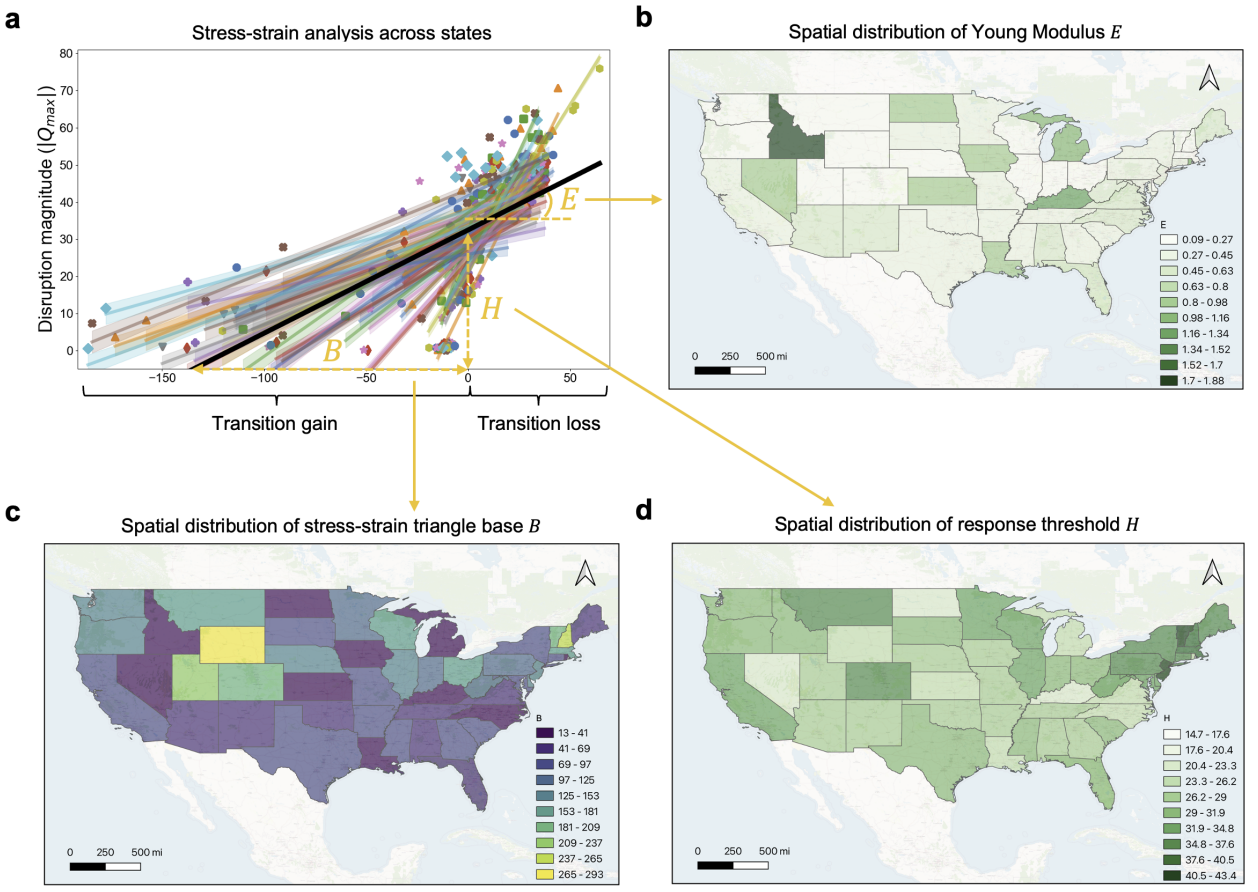


Figure 7: Resilience metrics analysis across states | a plots the regression curves with 95% confidence intervals for each state. **b**, **c**, and **d** show the spatial distribution of parameters E and response threshold H, and the metric of triangle base B across continent of the United States.

Table 3
Summary of System Parameters by Land Cover Type

Land Cover Type	System Base B	Young's Modulus E	R^2
Open Water	1.514	2578.718	0.957
Developed, Open Space	1.529	2639.548	0.966
Developed, Low Intensity	1.510	2632.552	0.971
Developed, Medium Intensity	1.454	2687.669	0.970
Developed, High Intensity	1.516	2617.858	0.970
Barren Land	1.517	2661.593	0.976
Evergreen Forest	1.840	1090.505	0.947
Mixed Forest	1.753	1249.769	0.911
Shrub-Scrub	1.591	2385.489	0.965
Grassland-Herbaceous	1.485	2652.155	0.979
Pasture & Hay	1.570	2596.775	0.975
Cultivated Crops	1.560	2657.390	0.980
Woody Wetlands	1.513	2756.112	0.972
Emergent Herbaceous Wetlands	1.559	2636.520	0.987
Forest (Combined Deciduous, Evergreen, Mixed)	1.840	1090.505	0.947
Developed (Combined Medium and High Intensity)	1.456	2688.320	0.969

Table 4
SSE comparison for model prediction of Transit Stations (TS) and Residential (RD) types in the United States

Mobility types	Models	SSE for H(t)	SSE for F(t)	SSE for h(t)	SSE for Δ
TS	LEA	0.191	0.029	0.037	0.002
	Exponential	2.100	0.249	0.044	0.024
	Log-logistic	1.556	0.193	0.042	0.018
	Weibull	3.084	0.342	0.046	0.031
RD	LEA	100.5	0.279	0.944	0.053
	Exponential	241.2	0.308	1.769	0.057
	Log-logistic	166.8	0.299	1.306	0.056
	Weibull	298.9	0.311	2.022	0.057

Figure 8a presents the recovery timeline of the transportation systems, illustrating the restoration progress of road, rail, airport, and seaport systems. These demonstrate similar patterns of early intensive recovery efforts, followed by a more gradual restoration phase. Figure 8b shows the recovery dynamics in percentages of water outages for a total of 2.3 million households and of Bullet train system following the earthquake. The estimated curve based on the LEA model is also shown simultaneously. Additionally, Figure 8c shows the similarly estimated $h(t)$. This figure demonstrates a rapid initial recovery and a peak in recovery capacity R_{cap} within the first week.

The Japan case is used as an external illustration of trajectory-level transferability. LEA curves and peak healing rates are estimated separately for the observed infrastructure recovery series. We do not construct a probabilistic network-reliability model because the available aggregate restoration records do not identify statistical dependence among infrastructure subsystems.

These applications illustrate the transferability of episode-based recovery analysis across mobility, thermal, and physical-infrastructure trajectories.

6. Discussion and Policy Implications for Resilient Urban Systems

Interpretive scope of the framework. Before stating policy-relevant implications, we restate the interpretive scope of the framework. The model presented in this paper is *descriptive and diagnostic*: it characterizes recovery trajectories observed in mobility, thermal, and infrastructure data, and quantifies case-specific diagnostic quantities (R_{cap}, E, B) that summarize how systems behave. The framework is not a structural causal model: it does not, by itself, identify the policy levers (e.g., targeted aid, infrastructure investments) that would change the recovery trajectory in a counterfactual sense. The policy implications discussed below should therefore be read as *questions for further evaluation*—e.g., via natural experiments, quasi-experimental designs, or controlled simulation studies—rather than as conclusions derived from causal evidence.

The proposed capacity-oriented quantification model introduces a framework that may support urban resilience analysis in the context of climate change and other disruptions. Using observed values of R_{cap} and B as diagnostic indicators, planners can identify regions and sectors whose recovery trajectories are atypical and warrant further investigation. Whether targeted interventions should be deployed in such regions—and what form they should take—requires complementary causal analysis and stakeholder engagement that this framework can inform but does not by itself supply. In particular, regions with low R_{cap} may exhibit slower observed recovery for multiple reasons (limited resources, structural composition of the local economy, exposure differences); distinguishing these explanations is a prerequisite to policy prescription.

The proposed indicators are diagnostic inputs to subsequent decision analysis. They may help identify episodes, sectors, or regions requiring further investigation, but they do not by themselves prescribe interventions or establish causal policy effects.

The present analysis has several limitations. The datasets used primarily reflect human mobility and large-scale recovery, leaving gaps in understanding how specific infrastructure types, such as water or energy systems, respond to disruptions. To generate a complete picture of urban resilience, more diverse datasets are needed that include non-residential buildings, varied infrastructure types, and social and economic factors. Broader datasets would be needed to assess whether the fitted relationships transfer across different urban environments.

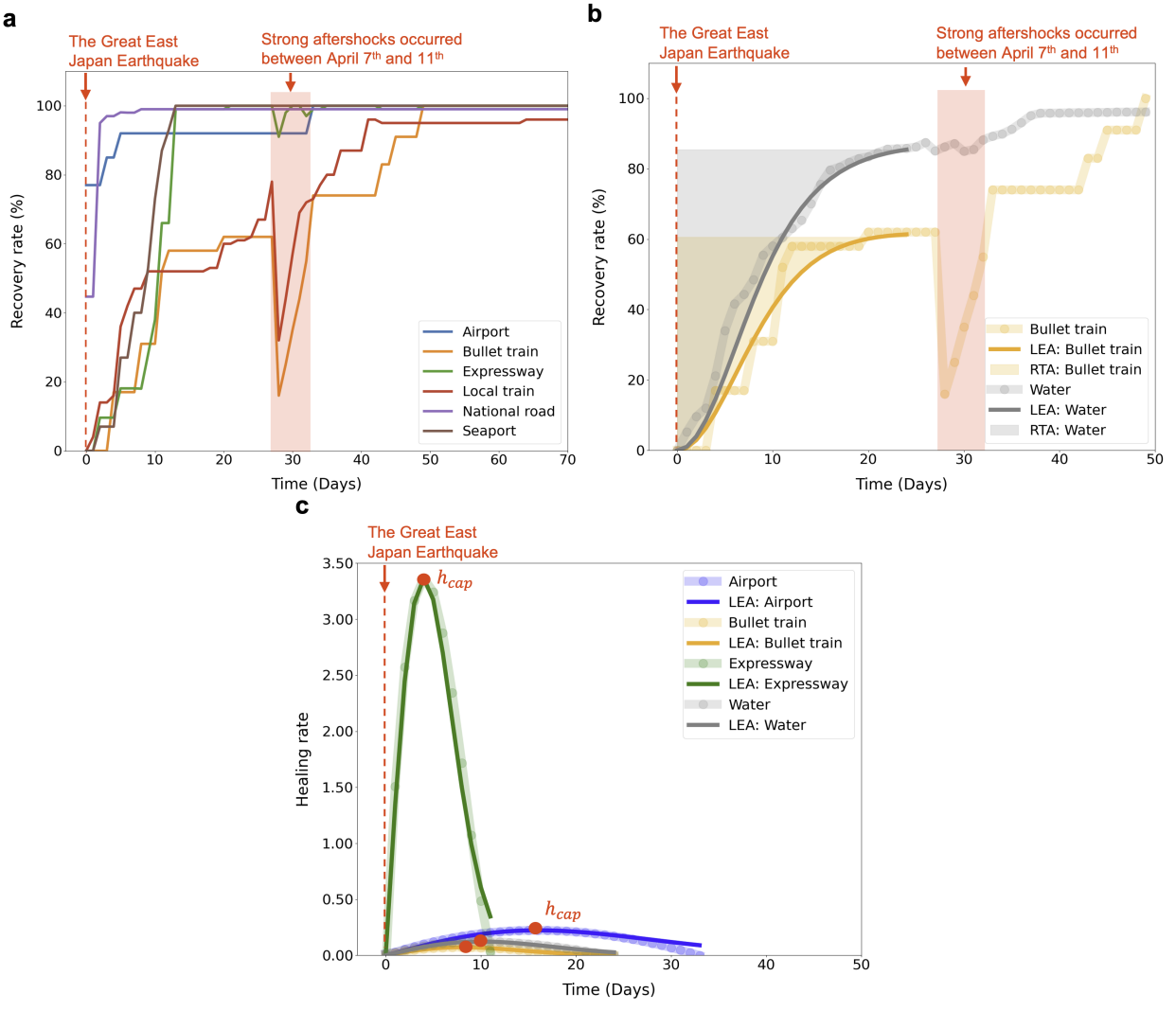


Figure 8: Recovery Dynamics in Japan Post-Earthquake. a $F(t)$ of transportation network recovery timeline illustrating the restoration progress of road, rail, airport, and seaport systems. **b** Observed data and $F(t)$ of water recovery timeline, showing the percentage of water restored over time. **c** Estimated $h(t)$ with a peak in recovery capacity R_{cap} of water recovery timeline in the early days following the earthquake.

Connection between Episode-Based and Multi-Episode Recovery Modeling: The Stress-Strain model represents each episode (disruption-recovery event) as a data point, where stress (Q_{max}, or total energy/disruption magnitude) and strain (e.g., recovery duration) are plotted. By aggregating multiple episodes across different communities or time periods, general trends can be observed. The LEA model, in contrast, focuses on individual recovery trajectories, tracking transition losses and gains over time. This trajectory-level perspective complements the cross-episode regression by separating recovery pace from the final transition loss or gain.

7. Concluding Remarks: From Transportation to Multi-Sector Infrastructure

Professor Hani S. Mahmassani's pioneering work demonstrated how rigorous theoretical frameworks grounded in traffic flow theory, optimization, and system dynamics could illuminate practical urban challenges. His approach—integrating demand, supply, behavioral adaptation, and network effects—established enduring principles that extend far beyond transportation systems.

The recovery capacity framework developed in this paper embodies the philosophy Professor Mahmassani championed: ground analysis in sound theory, develop models capturing essential dynamics, validate rigorously with data, and interpret results in relation to practical system questions. His work established that capacity is dynamic and state-dependent, that systems exhibit multiple operating regimes, and that behavioral adaptation and information shape time-dependent trajectories. These principles, originally developed for transportation networks, motivate the focused definition of recovery capacity examined in this paper.

A. Notation and Model-Comparison Summaries

A.1. Notation and Symbols

Table A1: Summary of Mathematical Notation and Symbols

Symbol	Definition
	Recovery Trajectory and Dynamics (following Tang et al. 2025)
$Q(t)$	Mobility pattern changes at time t
$Q_D(t)$	Disruption process
$Q_R(t)$	Recovery process
Q_{max}	Normalized peak disruption magnitude at t_2 (relative to baseline)
t_0, t_1, t_2, t_3, t_4	Critical time points (initial steady state, max gradient disruption, peak/valley, max gradient recovery, final steady state)
R	Total recovery duration, $R = t_4 - t_2$
Δ	Transition loss (negative) or transition gain (positive): $\Delta = Q(t_4) - Q(t_0)$
M	Normalization factor, $M = \max\{\lvert Q_{max}\rvert, \lvert Q_{max}\rvert + \Delta\}$
	LEA Model Functions
$h(t)$	Instantaneous healing rate (recovery capacity function)
$H(t)$	Cumulative healing capacity, $H(t) = \int_0^t h(\tau)d\tau$
$F(t)$	Forward trajectory function (recovery fraction), $F(t) = 1 - S(t)$
$S(t)$	Sub-remaining function (unrecovered proportion), $S(t) = e^{-H(t)}$
$f(t)$	Recovery speed function, $f(t) = h(t) \cdot S(t)$
a	Initial recovery speed parameter
b	Duration of recovery exploration phase (time of peak)
c	System's transition ability to steady state (decay factor)
ϵ	Small positive regularization value
	Recovery Capacity Metrics
R_{cap}	Single-period recovery capacity (maximum healing rate); $R_{cap} = a \cdot b \cdot e^{-c}$
RTA	Recovery/Resilience Trajectory Area
	Stress-Strain Model
σ, ε	System stress (disruption magnitude) and strain (relative duration/deformation)
E	Magnitude of the fitted approximately linear stress–strain slope, $E = -d\sigma/d\varepsilon$
H	System response threshold (intercept in stress-strain relation)
B	Stress-strain triangle base (resilience metric), $B = H/E$
	Extreme-Heat Episode Indicators
$\lambda(t)$	Observed solar-radiation inflow (GHI)
D	Cumulative solar loading, $D = \int_{t_0}^{t_3} \lambda(\tau)d\tau$
P	Post-peak extreme-heat duration, $P = t_3 - t_2$

A.2. SSE for Mobility Types Across 50 States

The following table summarizes the SSE for different mobility types across 50 states during model fitting and prediction phases.

Table A2
SSE of 50 states for different mobility types in fitting and prediction

Mobility types	Models	Combined SSE for model fitting	SSE for F(t) prediction	SSE for Δ
GP	LEA	0.015	0.227	0.515
	Exponential	0.032	0.688	2.195
	Log-logistic	0.076	0.497	1.723
	Weibull	0.132	0.556	2.105
PK	LEA	0.091	0.996	2.081
	Exponential	0.071	1.017	2.311
	Log-logistic	0.222	1.041	2.508
	Weibull	0.139	1.098	2.932
RD	LEA	0.532	3.680	10.169
	Exponential	0.896	5.659	15.157
	Log-logistic	1.839	4.397	12.170
	Weibull	1.749	3.971	11.307
RR	LEA	0.004	0.243	0.536
	Exponential	0.010	0.332	1.126
	Log-logistic	0.019	0.409	1.046
	Weibull	0.047	0.253	0.896
TS	LEA	0.006	0.134	0.519
	Exponential	0.008	0.090	0.788
	Log-logistic	0.025	0.039	1.700
	Weibull	0.042	2.295	0.790
WP	LEA	0.003	0.152	0.272
	Exponential	0.005	0.349	0.851
	Log-logistic	0.029	2.961	4.838
	Weibull	0.034	0.269	0.668

Code and Data Availability

The code implementing the LEA model, the DQQ episode filter, and the stress–strain calibration, together with the processed data used to reproduce the figures and tables, is available at `https://github.com/asu-trans-ai-lab/recovery-capacity`. The Google COVID-19 Community Mobility Reports, NLCD, and NSRDB source data are publicly available from their original providers; additional processed data are available from the corresponding authors upon request.

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